

FLEXHAND · REQUIREMENTS

FlexHand requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/flexhand		
Source	docs/03-requirements.md @ abb956e		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlexHand requirements

These are first-pass requirements for the concept. Targets are proposals for review with a therapist and will be checked by calculation at TRL 3. The status column gives the TRL 2 estimate from the design precis (FXH-PRC-001); two requirements are not met and two are at risk.

Table 1. Requirements and concept status.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R1	Move the four fingers through a useful range	MCP 0 to 70°, PIP 0 to 90°, within therapist-set limits	Tendon excursion calculation; later bench test on a finger model	Met by design: about 25 mm tendon excursion needed, 35 mm provided
R2	Extend fingers against moderate flexor tone	30 N extensor tendon force per finger at the finger	Torque and friction calculation	Met, thin margin: 0.41 N·m at the spool, about 5 % above the gearbox's recommended continuous load
R3	Limit force on the hand	Software limit 40 N per finger from motor current; mechanical breakaway at 60 N or less per tendon	Calculation; later bench test	By design, unverified
R4	Deliver a high repetition dose	300 or more full flexion and extension cycles in a 60 min session	Speed calculation	Met: about 6.4 cycles per min, about 385 per hour at design load
R5	Move slowly and smoothly	Stroke time adjustable from 3 to 15 s (joint speed about 6 to 30 °/s); dwell at each end adjustable	Firmware sketch and speed calculation	Met by design
R6	Run several sessions per charge	3 or more 60 min sessions at design load	Power budget	Met, no margin: about 3.1 sessions
R7	Light enough to wear seated	Hand-side parts 120 g or less; forearm pack 450 g or less	Mass estimate, then weighing	Not met: hand about 90 g (met), pack about 510 g
R8	Quick to put on and take off	A care partner dons it in 5 min or less and removes it in 1 min or less	Design review with users	At risk: tendon cuffs on flexed, spastic fingers may be hard to fit
R9	Stop and release on demand	Physical stop cuts motor power within 100 ms; tendons released by hand in 10 s or less without tools	Design review; later bench test	By design, unverified
R10	Fit most adult hands	Hand length 170 to 205 mm with three glove sizes and adjustable cuffs	Anthropometric check	By design
R11	Keep contact pressure tolerable	Mean pressure under a finger cuff 50 kPa or less at design load (proposed limit, to be set with a therapist)	Contact area calculation	Not met: about 100 kPa with 12 mm cuffs
R12	Low cost and buildable	Parts cost \$500 or less; no custom PCB for the first build	Priced BOM	Met: about \$280
R13	Record what was done	Device logs session time, cycle count and peak motor current; therapist limits cannot be changed from the user controls	Design review	By design

■ Assumptions

- Design load: a finger with moderate flexor tone needs about 30 N of extensor tendon force to reach full extension at the set speed. This is an estimate consistent with the Columbia orthosis, whose single motor provides about 100 N peak for four fingers ([Park et al., arXiv 1802.06131](#)). It must be checked against published finger stiffness data at TRL 3.
- Tendon excursion for 70° at the MCP and 90° at the PIP is about 25 mm, taking dorsal moment arms of about 10 mm and 8 mm; 35 mm is allowed for cuff compliance and slack.
- Bowden sheath efficiency is about 73 %, from the capstan relation with a friction coefficient of 0.1 and a total bend of 180° (π rad).
- The forearm rests on a support during sessions. The device is not designed to be worn while walking.
- Sessions are supervised by a care partner, at least until a therapist agrees otherwise.

Safety: R3, R9 and R11 are safety requirements. A design that misses any of them must not be worn by a person, even at TRL 4 bench stage.